

Nathan Cheng

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Education

University of South Florida

B.S. Computer Science

Aug 2021 – Dec 2025

Experience

Microsoft

Software Engineer

Apr 2026 – Present

- Automated internal release engineering workflows at Office Products using AI tools and MCP servers, reducing engineering task workloads by up to 80%.
- Architect interactive frontend web applications and telemetry dashboards using TypeScript and React to monitor live system metrics.
- Engineered a Responsible AI agent to audit model inputs, enforce safety protocols, and streamline AI compliance workflows for cross-functional teams.

Apple

Software Engineer Intern

May 2024 – Aug 2024

- Engineered user-facing media playback and spatial video features for Apple Vision Pro platform applications using Swift.
- Reduced video streaming latency by 40% through asynchronous pipeline optimizations and low-level frame buffering techniques.
- Collaborated with UI/UX engineers to implement fluid, high-frame-rate spatial UI components into core OS infrastructure.

MIT Lincoln Laboratory

Research Intern

Jan 2024 – Apr 2024

- Developed an end-to-end computer vision and video processing pipeline in Python and OpenCV, accelerating model processing speeds by 80%.
- Implemented custom image processing algorithms for spatial feature extraction and automated media data transformation.

Projects

USublease

Oct 2025

- Architected a full-stack university sublease marketplace serving 5,000+ users with a responsive React/TypeScript frontend and a Python (FastAPI) backend containerized on Google Cloud Run.
- Designed Firestore database schemas and Google Cloud Storage (GCS) pipelines to handle real-time geospatial property filtering, dynamic image processing, and Mapbox integrations.
- Engineered core marketplace features including Stripe webhooks for monetization, OAuth authentication, and an automated Cloud Build multi-environment CI/CD deployment pipeline.

Technical Skills

Languages: Python, TypeScript, JavaScript, Swift, C/C++, SQL, HTML/CSS

Frontend & Web: React, Frontend Architecture, UI/UX Integration, Tailwind CSS, HTML5/CSS3

Backend & Libraries: FastAPI, Node.js, OpenCV, REST APIs, Webhooks

Cloud & Tools: Google Cloud Platform (GCP), Docker, Git, Firestore, Mapbox, MCP Servers, CI/CD